` **COURSE CODE**

SBT-DF204

COURSE TITLE

Computer Forensics Case Studies

LAB TITLE

Investigating Harassment Email Traffic with Wireshark

NAME

Kumpon Emmanuel Nandom

STUDENT ID

2025/FWSD/11325

DATE

5/10/2026

PLATFORM USED

Kali Linux

INSTRUCTOR

Mr. Aminu Idris, AMCPN

Commandant

Case reference and student identification

Student name	Kumpon Emmanuel
Registration number	2025/FWSD/11325
Course code	SBT-DF204 Computer Forensics Case Studies
Case title	Investigating Harassment Email Traffic with Wireshark
Evidence file	nitroba.pcap (working copy nitroba_working.pcap)
Analysis platform	Kali Linux, Wireshark; analysis performed 4 to 5 October 2026
Time convention	All absolute times are UTC.

Executive Report

The packet capture shows that one device on the residence network, identified by MAC address 00:17:f2:e2:c0:ce and IP address 192.168.15.4, loaded the form page of www.willselfdestruct.com (69.25.94.22) and then submitted an HTTP POST. The posted form is addressed to lilytuckrige@yahoo.com, carries the subject "you can't find us", and contains the message text reported by the complainant. The server accepted the submission.

The same device also sent traffic to mail.google.com carrying a cookie that includes the Gmail address jcoachj@gmail.com. That account name is consistent with the roster entry "Johnny Coach" in Chemistry 109, but the roster contains no email addresses, so the link is an inference.

The evidence supports the device and the Gmail account being active on it, as a credible lead to a Chem 109 student. It does not establish who was operating the device at the time, because the residence uses an open, shared wireless network and the capture cannot show who was at the keyboard. Confidence is high for the device-level finding, moderate for the device-to-account link, and low to moderate for naming an individual.

Evidence acquisition and integrity

The traffic capture published with the Nitroba University Harassment Scenario was downloaded to a dedicated working folder SBT-DF204-Lab1 on Kali Linux. A bit-identical working copy was created with cp -p, and the original was set to read-only (chmod 444) so that all analysis used the working copy.

Item	Recorded value
Original file name	nitroba.pcap
Source	https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
File size	56,180,821 bytes
Download date	4 October 2026 9:27 pm
SHA-256 (original)	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA-256 (working copy)	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb, identical to the original
MD5 (cross-reference)	9981827f11968773ff815e39f5458ec8.
Working copy method	cp -p nitroba.pcap nitroba_working.pcap; sha256sum run on both files; original set to mode 444
Published checksum comparison	CONFIRM

Method

The capture was opened in Wireshark (94,410 packets). Because the capture is large, each step used a display filter to narrow the traffic, and each hit was verified in the packet details before being treated as a finding.

Step	Display filter	Purpose and result	Figures
1	http.request.method == "GET" && http.host == "www.willselfdestruct.com"	Locate web requests to the service. 9 packets, all from 192.168.15.4 to 69.25.94.22.	Fig 5
2	data-text-lines contains "secure anonymous E-mail"	Find server responses by page title (field name found via Prepare as Filter). 2 packets, 82998 and 83641.	Figs 6 to 8, 15, 16
3	(ip.src == 192.168.15.4) && (ip.dst == 69.25.94.22) && http.request.method == "POST"	Isolate the form submission. 1 packet, 83601.	Figs 9 to 12

Step	Display filter	Purpose and result	Figures
4	eth.addr == 00:17:f2:e2:c0:ce	Trace the device. 73,246 of 94,410 packets (77.6%).	Fig 13
5	eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"	Look for identity evidence in cookies. 149 packets.	Fig 14
6	Follow TCP Stream on packet 83601 (filter tcp.stream eq N, where N is the TCP stream index in the TCP header of packet 83601)	Review the complete POST exchange as one stream.	Fig 11
7	View > Time Display Format > UTC Date and Time of Day	Convert relative timestamps to absolute UTC for the timeline.	Figs 7, 8, 10

Findings

Q1. What file was acquired and how was its integrity preserved?

The file is nitroba.pcap, 56,180,821 bytes, with SHA-2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb. A working copy was made with cp -p, shown to have the same SHA-256 hash, and the original was made read-only. The MD5 also agrees with the reference value in the course material.

Q2. Which client system contacted the web service?

The Packet is an HTTP GET for /secure/submit with Host: www.willselfdestruct.com, sent from 192.168.15.4 to 69.25.94.22. The server replied with HTTP 200 OK, from 69.25.94.22 to 192.168.15.4, later. The page title "FREE secure anonymous E-mail to a friend, client or colleague: WillSelfDestruct" matches the data-text-lines filter.

Q3. What evidence links the client to the harassment message?

The Packet is an HTTP POST to /secure/submit, from 192.168.15.4 to 69.25.94.22. Wireshark lists an Ethernet stream.

Form item	Value
to	lilytuckrige@yahoo.com
from	-
subject	You can't find us
message	And you can't hide from us. / Stop teaching. / Start running.

This corresponds to the second harassment email as received and displayed in the scenario material. The subject and body text are the same; the empty "from" field corresponds to what is

shown on the service page and corresponds to the notice that the message could be viewed once for 30 seconds.

Q4. Which device made the relevant request?
In the Ethernet II header of the packet, the source MAC address is 00:17:f2:e2:c0:ce, which Wireshark resolves to Apple_e2:c0:ce. The destination MAC is 00:1d:d9:2e:4f:60. Filtering on eth.addr == 00:17:f2:e2:c0:ce returns 73,246 of the 94,410, so this device accounts for most of the traffic in the capture.

Why the IP address alone is not enough. The scenario describes a shared student room with three occupants and a Wi-Fi router with no password. An IP address in that environment identifies a connection, not a person: any occupant, a visitor, or anyone within wireless range could have used it, and several devices can share one address. The MAC address narrows the question to one physical device, but it still identifies a device rather than its user, and MAC addresses can be changed or spoofed.

Q5. Who is associated with the device?
Filtering for packets from the same MAC with cookies containing "@" returned 149 packets. In the packet, a GET to mail.google.com, the Cookie header includes the pair gmailchat=jcoachj@gmail.com/475090.

A browser on this device had an active Gmail session for the account jcoachj@gmail.com. The capture does not show a login, so it does not show who typed credentials, and it does not show who was using the device at the moment of the POST. Packet 80824 occurred about 82 seconds before the POST.

Q6. Is that person on the Chem 109 roster?

No.	Roster entry	Comparison with cookie evidence
1	Amy Smith	No match
2	Burt Greedom	No match
3	Tuck Gorge	No match
4	Ava Book	No match
5	Johnny Coach	Account name "jcoachj" is consistent with first name initial J, surname Coach.
6	Jeremy Ledvkin	No match
7	Nancy Colburne	No match
8	Tamara Perkins	No match
9	Esther Pringle	No match
10	Asar Misrad	No match
11	Jenny Kant	No match

The roster comes from the scenario material. Only one entry shares the surname in the account name. This is a pattern match, not a confirmed identity: the roster has no email addresses, and the evidence does not show that the account belongs to that student rather than to someone else with a similar name.

Q7. When did the activity occur?
Wireshark was first showing relative time. After changing to View > Time Display Format > UTC Date and Time of Day, packets 82998 and 83641 display as 2008-07-22 06:03:44.210946Z and 2008-07-22 06:04:24.851431Z. The time zone displayed is therefore UTC. Other times in the table are derived by subtracting relative-time differences from these two anchors and are marked with a dagger.

UTC, 22 July 2008	Relative (s)	Packet	Event
06:03:02.386	15115.291	80824	Device sends Gmail request with Gmail chat cookie
06:03:43.826	15156.731	82936	GET /secure/submit (form page requested)
06:03:44.211	15157.116	82998	Server returns form page (HTTP 200)
06:04:24.312	15197.216	83601	POST with harassment message
06:04:24.469	15197.374	83604	Server returns HTTP 302
06:04:24.564	15197.469	83614	GET /secure/success
06:04:24.851	15197.756	83641	Server returns success page (HTTP 200)

Q8. What conclusion can be defended?
The capture directly shows that the device with MAC 00:17:f2:e2:c0:ce submitted the harassment message through willselfdestruct.com, and it links that device to the Gmail account jcoachj@gmail.com. It does not identify who used the device.

Timeline

Time	Event
13 Jul 2008, 17:21:01 (-0700)	First harassing email received
22 Jul 2008, 06:03:43.826	Client requests the willselfdestruct.com form page
22 Jul 2008, 06:04:24.312	Client posts the harassment message
21 Jul 2008, 11:04 PM	Second email appears in the complainant's Yahoo inbox; equals 06:04 UTC if the mailbox used PDT
22 Jul 2008, 06:04:24.851	Success page returned to the client

Attribution assessment

1. Device 00:17:f2:e2:c0:ce / 192.168.15.4 requested the service page at packet.
2. A packet from the same device posted a form to lilytuckrige@yahoo.com with the subject and message text reported by the complainant.
3. The server redirected the client to a success page.
4. The same MAC sent 149 packets with "@" cookies, including gmailchat=jcoachj@gmail.com/475090.

Confidence level

Statement	Confidence	Basis
The device 00:17:f2:e2:c0:ce submitted the harassment message through willselfdestruct.com	High	Message content, IP, MAC and server response all appear in the capture.
That device was associated with the Gmail account jcoachj@gmail.com	Moderate	Cookie in 149 packets from the same MAC, login and timing not shown
The device user was the roster student Johnny Coach	Low to moderate	Name pattern only; open shared network, no direct evidence of the user.

Evidence log

ID	Packet or item	Finding	Why it matters	Figure
E01	Evidence file	nitroba.pcap, 56,180,821 bytes; SHA-2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb Identical for original and working copy; MD5 9981827f11968773ff815e39f5458ec8.	Identifies the exact file and shows it was preserved	Figs 1, 2, 3
E02	Whole capture	94,410 packets, unfiltered view	Establishes the scope before filtering	Fig 4
E03	Pkt 82936	GET /secure/submit, Host www.willselfdestruct.com, 192.168.15.4:35984 to 69.25.94.22:80.	Client first contacts the service; gives client and service IPs	Fig 5
E04	Pkt 82998	HTTP 200 from 69.25.94.22; title match "secure anonymous E-mail"; 06:03:44.211Z; 385 ms after request	Confirms service IP and client IP from the response side	Figs 6, 7
E05	Pkt 83601	POST /secure/submit, 192.168.15.4:36044 to 69.25.94.22:80; to lilytuckrige@yahoo.com; subject "you can't find us"; message text.	Content matches the reported harassment email	Figs 9, 10
E06	Pkt 83601 Ethernet header	Src MAC 00:17:f2:e2:c0:ce (Apple_e2:c0:ce); dst 00:1d:d9:2e:4f:60. Ethernet stream index 19	Identifies the device that sent the POST	Fig 12
E07	eth.addr filter	73,246 of 94,410 packets involve this MAC	Shows the device is the dominant client in the capture	Fig 13
E08	Pkts 83604, 83614	HTTP 302 then GET /secure/success.	Server accepted the submission	Figs 5, 13
E09	Pkt 83641	HTTP 200 for /secure/success; 06:04:24.851Z; 287 ms after request	Completes the exchange	Figs 6, 8

ID	Packet or item	Finding	Why it matters	Figure
E10	Pkt 80824 (1 of 149)	Same MAC, GET mail.google.com, Cookie gmailchat=jcoachj@gmail.com/475090; rel. 15115.291 s (06:03:02.386Z †)	Only identity evidence in the capture	Fig 14
E11	Chem 109 roster	Roster lists Johnny Coach; no email addresses given	Any roster link is an inference	Section 4, Q6 (roster table)
E12	Pkts 82998, 83641 headers	Server Date: 07:24:05 and 07:24:45 GMT vs capture 06:03:44Z and 06:04:24Z	Clock discrepancy to disclose as a limitation	Figs 7, 8
E13	TCP stream of pkt 83601 (index N to confirm)	Follow TCP Stream of the POST exchange	Shows the full request and response as one conversation	Fig 11 [to insert]
E14	Method	data-text-lines field identified via Prepare as Filter; 744 packets contain line-based text data	Shows how the title-search filter was built	Figs 15, 16

Labelled screenshots

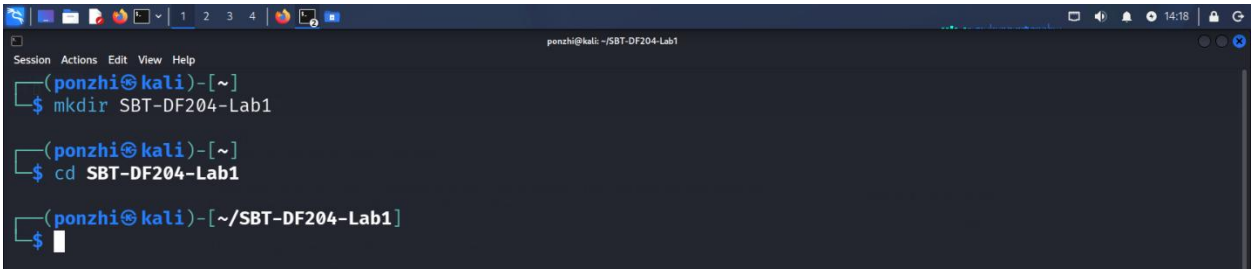


Figure 1. Working folder SBT-DF204-Lab1 created on Kali.

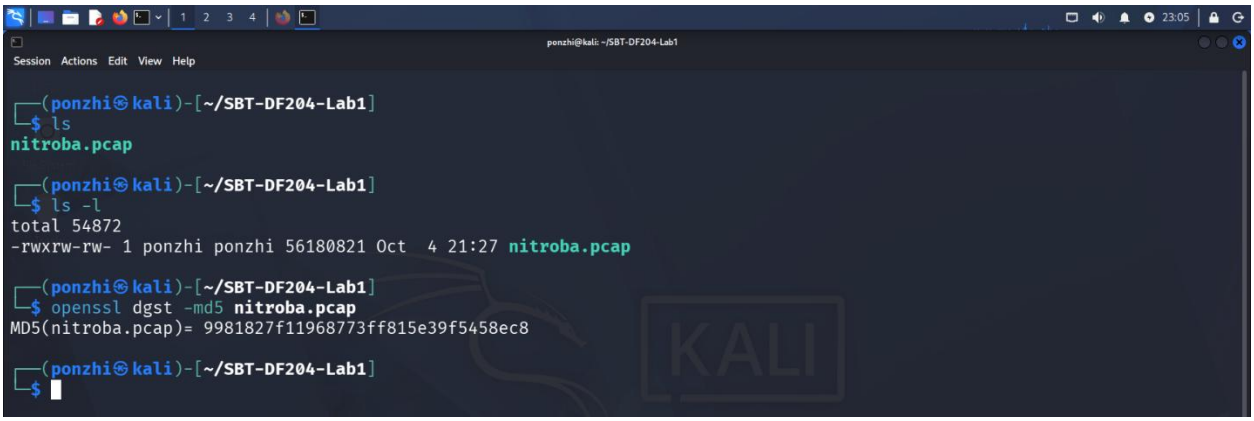


Figure 2. Original file listing (56,180,821 bytes, file time Oct 4 21:27) and MD5 value.



Figure 3. SHA-256 of nitroba.pcap and nitroba_working.pcap (identical), working copy made with cp -p, original set to chmod 444.

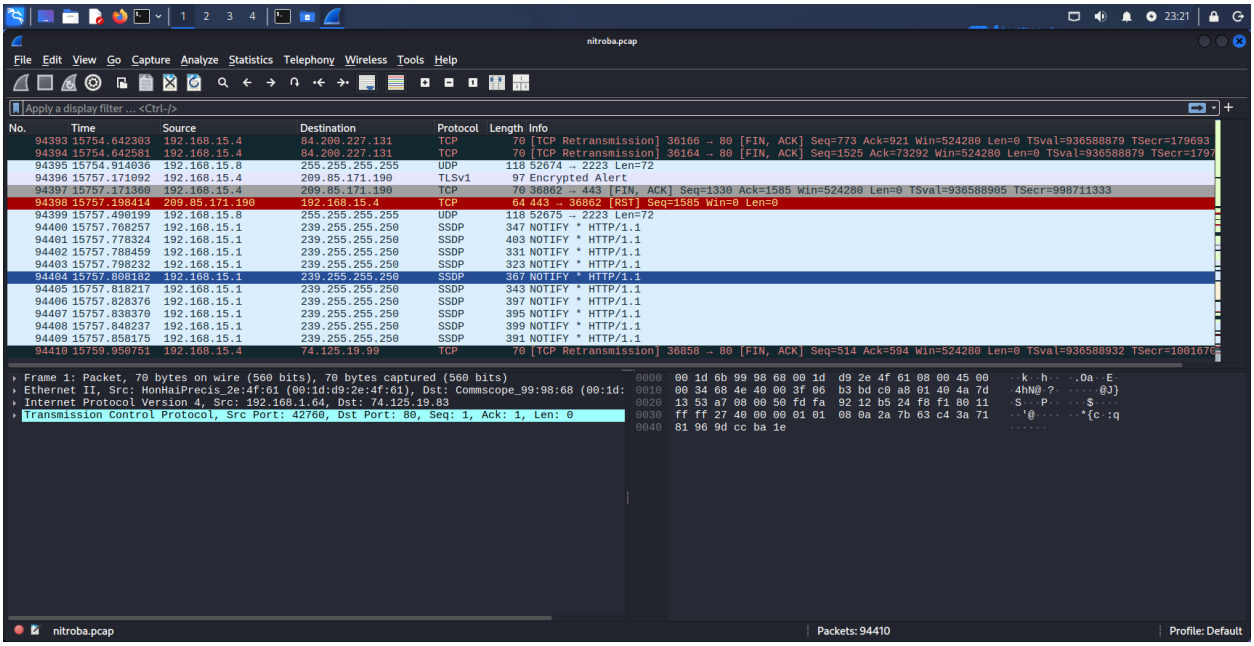


Figure 4. Capture opened with no filter: 94,410 packets.

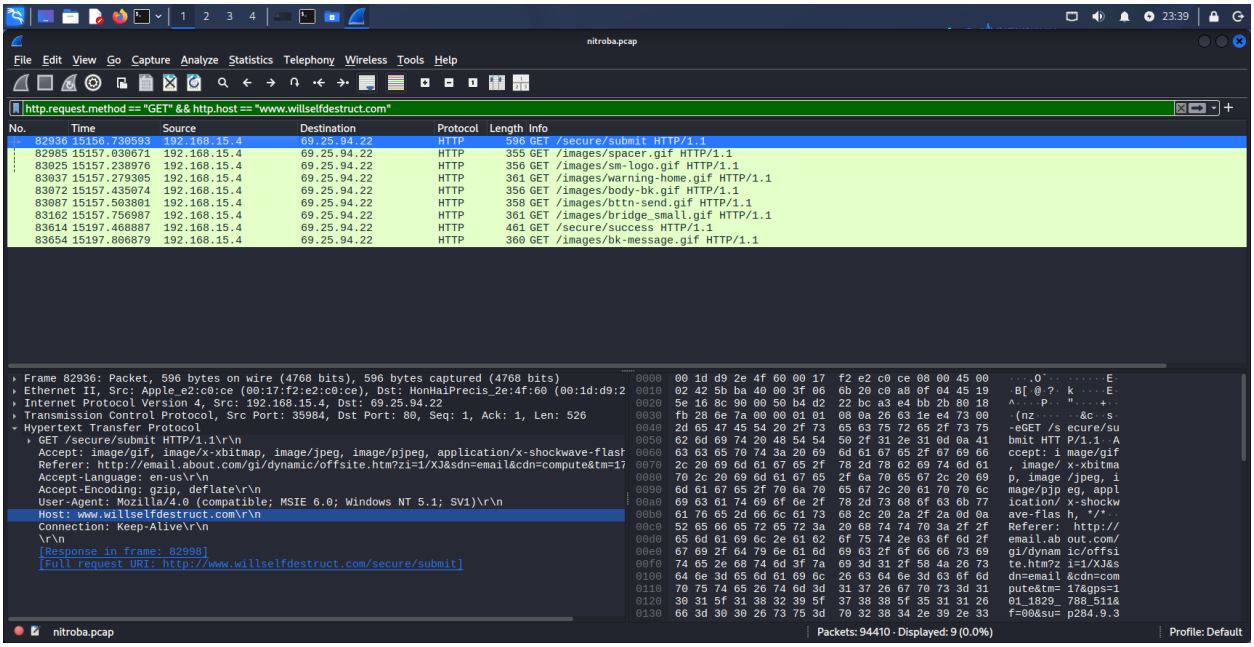


Figure 5. *Filter http.request.method == "GET" && http.host == "www.willselfdestruct.com": 9 packets; packet 82936 detail shows client 192.168.15.4, server 69.25.94.22, Host header and Response in frame 82998.*

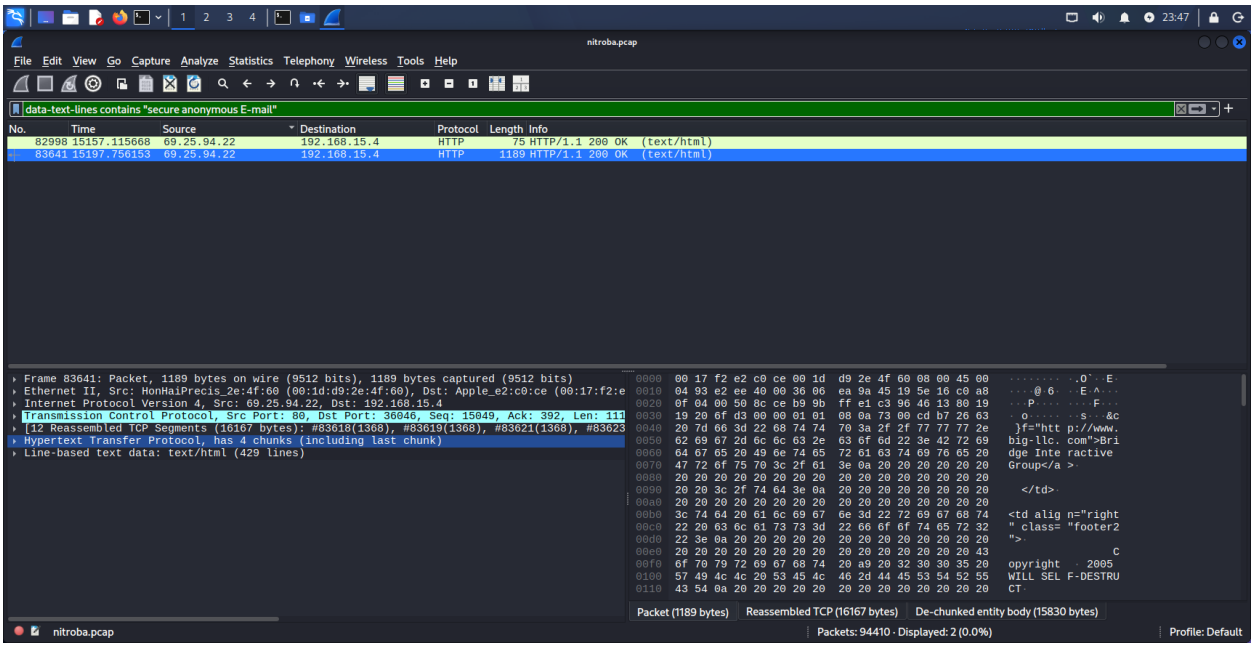


Figure 6. Filter data-text-lines contains "secure anonymous E-mail": responses 82998 and 83641 from 69.25.94.22 to 192.168.15.4.

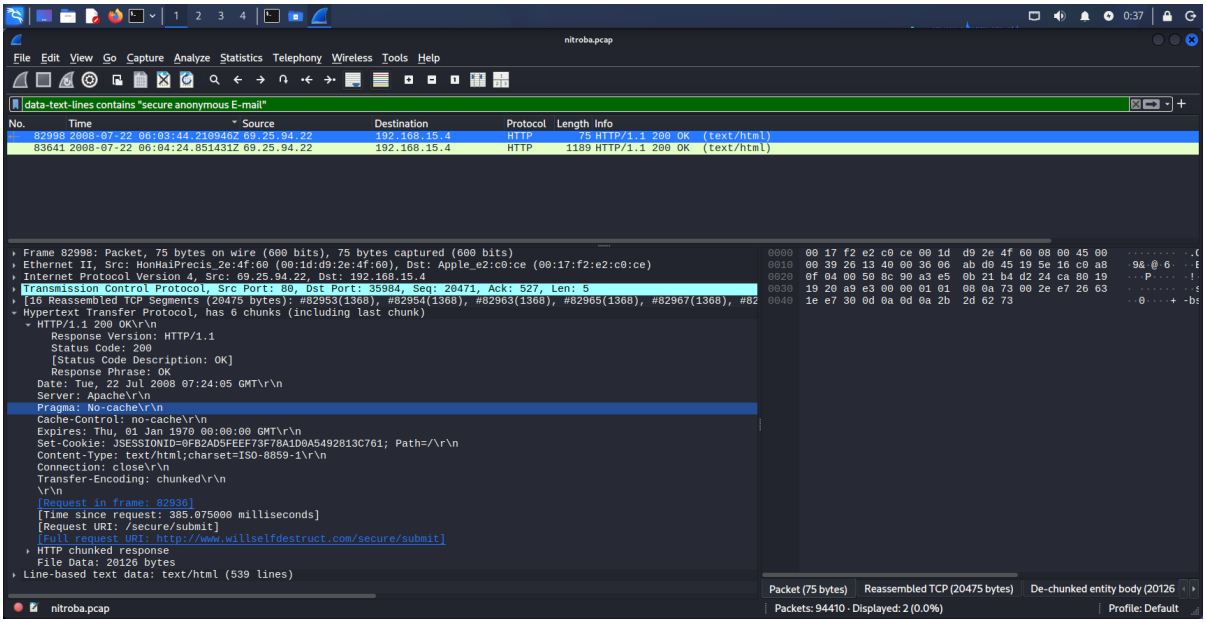


Figure 7. Packet 82998 with UTC time display: 2008-07-22 06:03:44.210946Z; response to /secure/submit in 385 ms; server Date header 07:24:05 GMT.

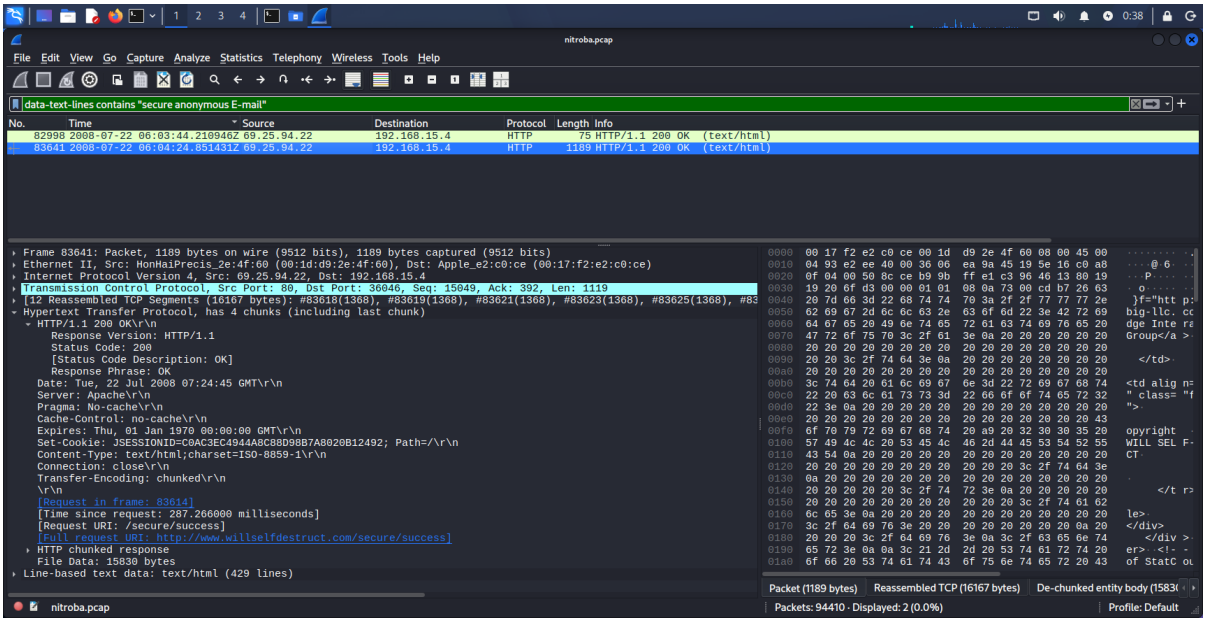


Figure 8. Packet 83641 with UTC time display: 2008-07-22 06:04:24.851431Z; response to /secure/success in 287 ms; server Date header 07:24:45 GMT.

Message correlation

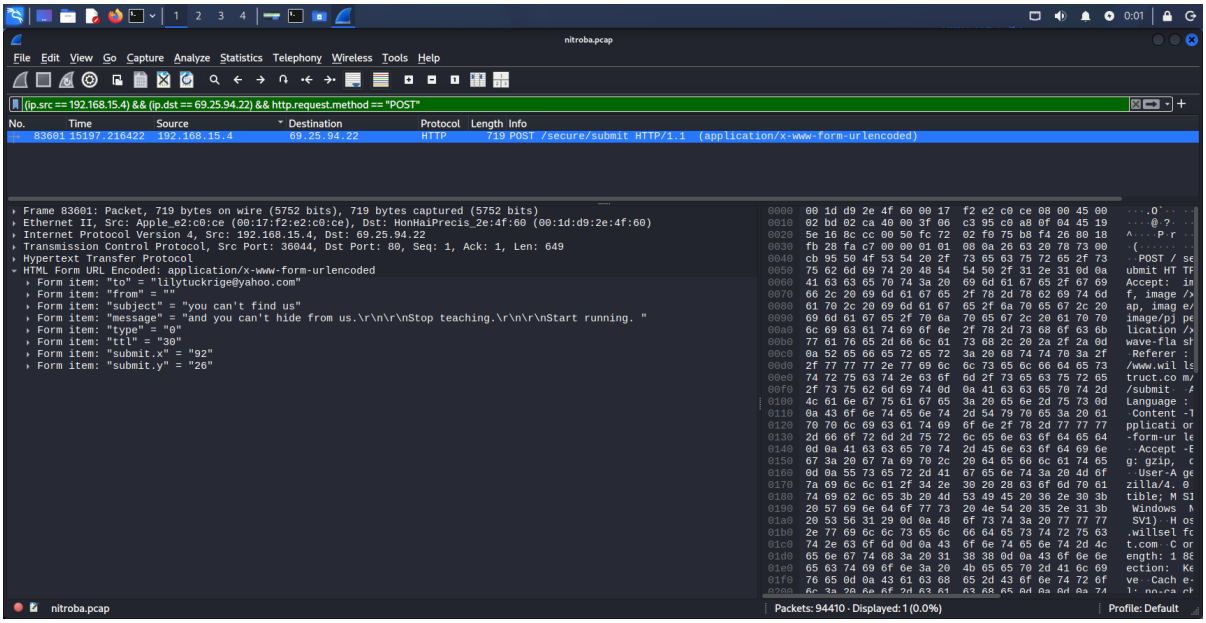


Figure 9. Filter (ip.src == 192.168.15.4) && (ip.dst == 69.25.94.22) && http.request.method == "POST": packet 83601 with form fields to, from, subject and message.

Device

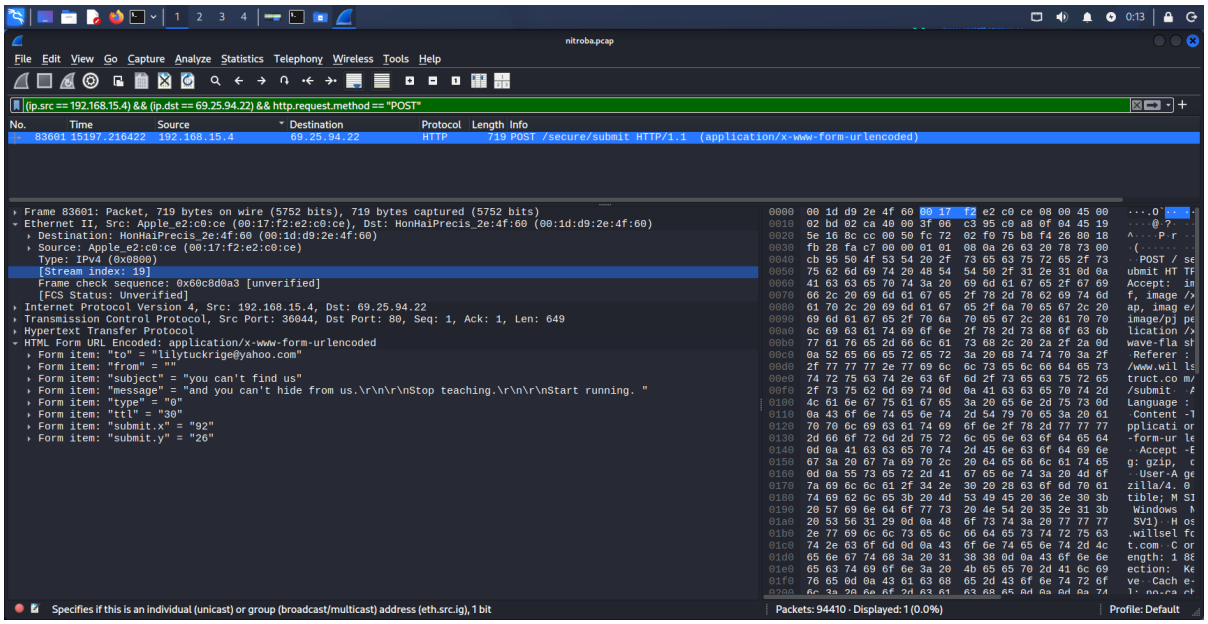


Figure 12. Packet 83601 Ethernet II header expanded: source MAC Apple_e2:c0:ce (00:17:f2:e2:c0:ce), destination HonHaiPrecis_2e:4f:60; Ethernet stream index 19.

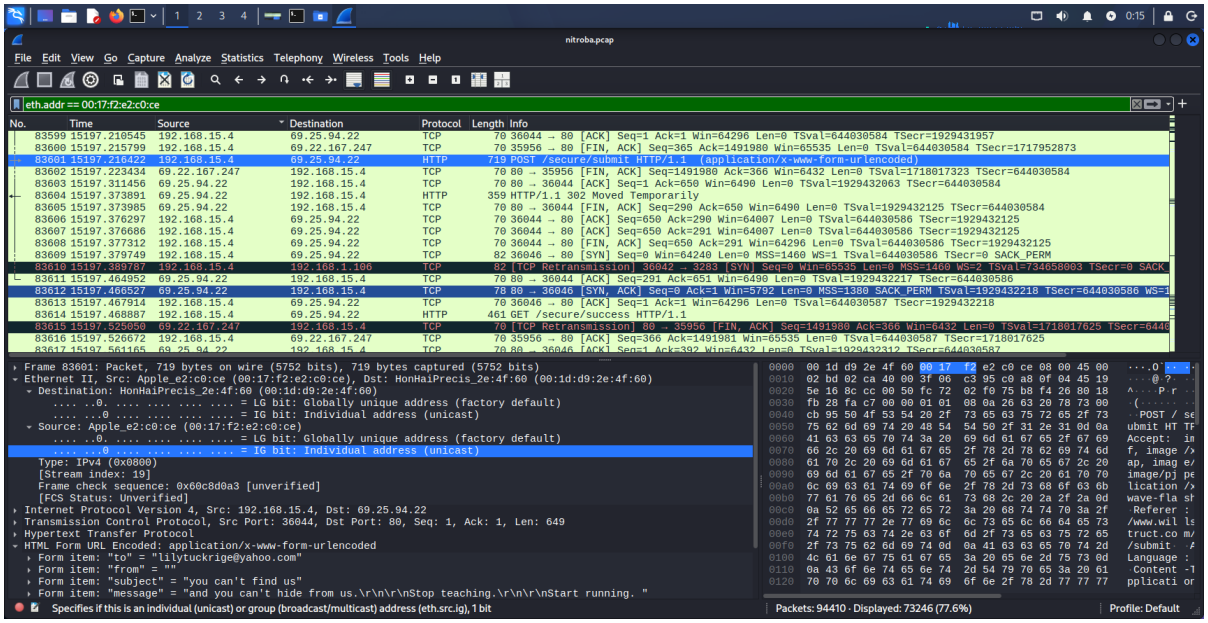


Figure 13. Filter eth.addr == 00:17:f2:e2:c0:ce: 73,246 of 94,410 packets (77.6%), including POST 83601, HTTP 302 (83604) and GET /secure/success (83614).

Identity evidence

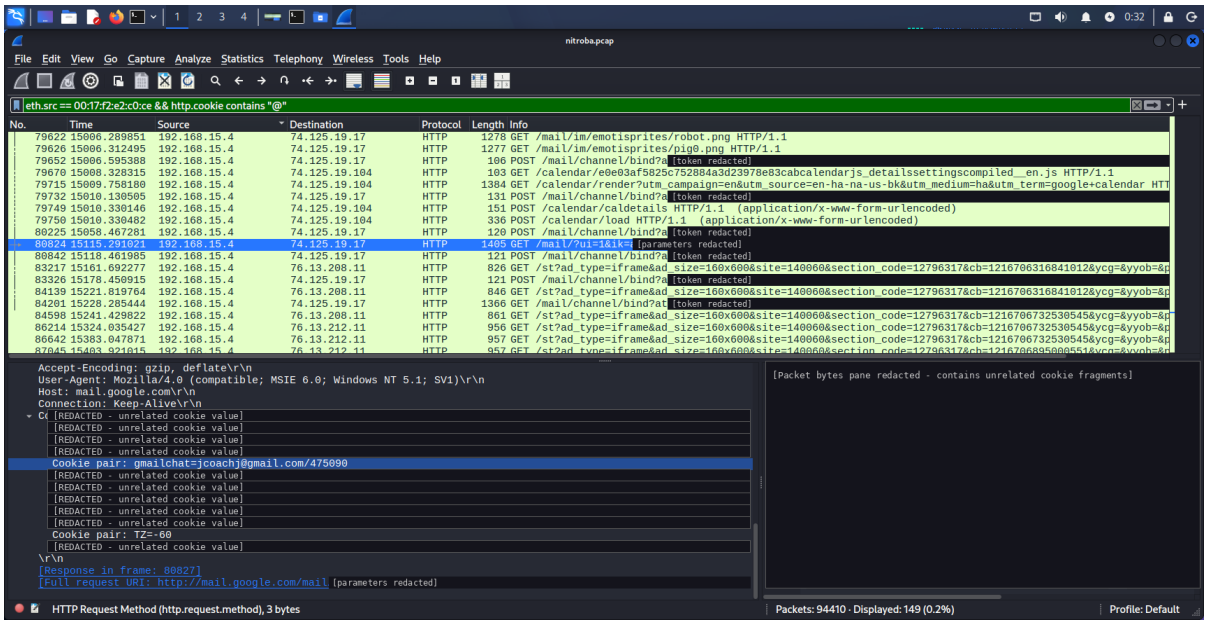


Figure 14. Filter eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@": 149 packets; packet 80824 shows Cookie pair gmailchat=jcoachj@gmail.com/475090. Unrelated cookie values and session tokens are redacted.

Method support

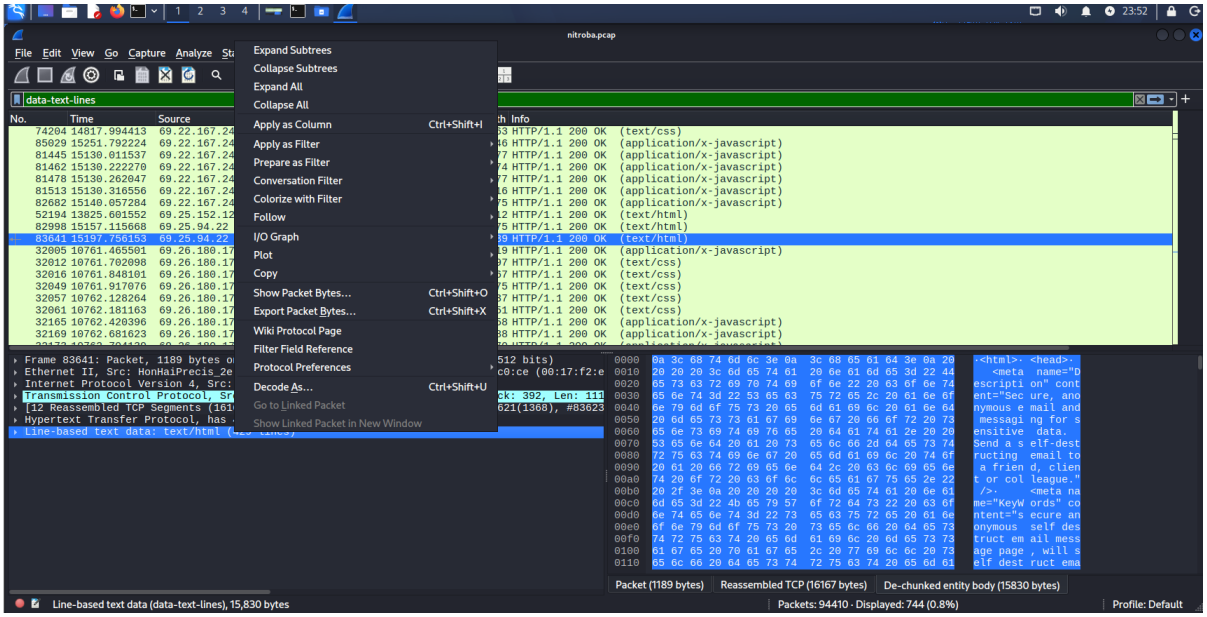


Figure 15. Finding the field name data-text-lines via right-click on Line-based text data > Prepare as Filter.

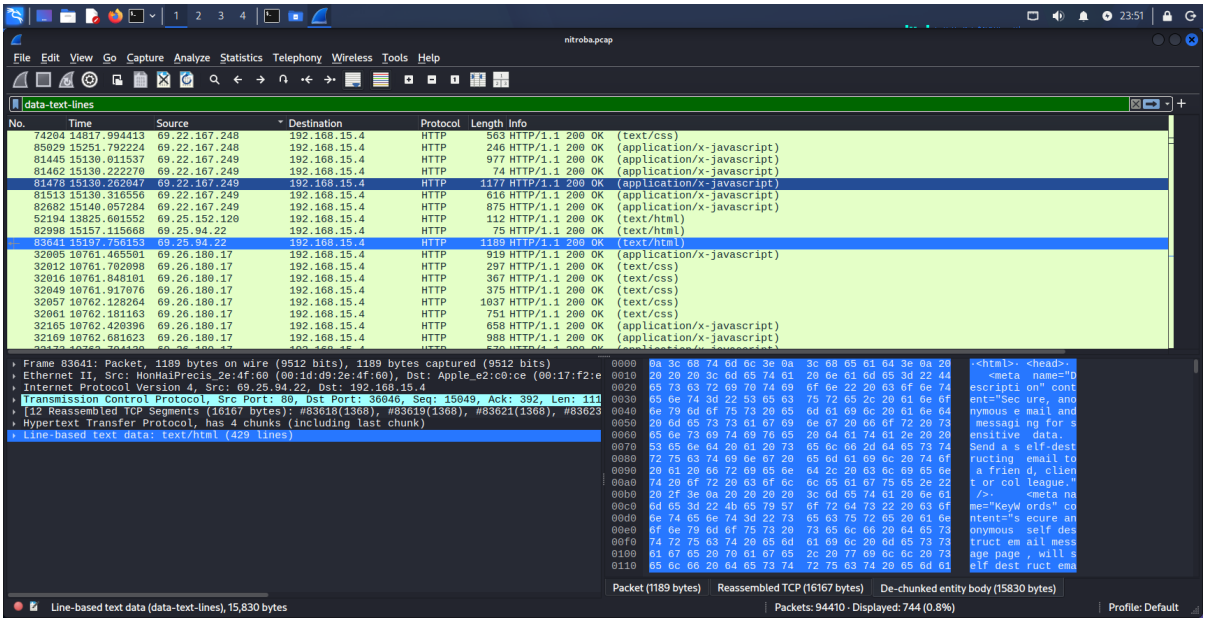


Figure 16. *Filter data-text-lines applied: 744 packets contain line-based text data.*

Appendix C. Command log

Command	Purpose
mkdir SBT-DF204-Lab1; cd SBT-DF204-Lab1	Create the working folder
ls; ls -l	Confirm file name and size (56,180,821 bytes)
openssl dgst -md5 nitroba.pcap	MD5 cross-reference with course material
sha256sum nitroba.pcap	SHA-256 of the original
cp -p nitroba.pcap nitroba_working.pcap	Create the forensic working copy
sha256sum nitroba_working.pcap	Verify the copy agrees with the original
chmod 444 nitroba.pcap	Make the original read-only